

ATM Cash Utilization & Downtime Analysis

Data Analytics Project Report

Python • Pandas • SQL • MySQL • Power BI

Project Objective

This project analyzes ATM transaction activity, cash demand, utilization, location patterns, time-based behavior and potential cash availability risk using a structured analytical dataset.

1. Dataset Overview

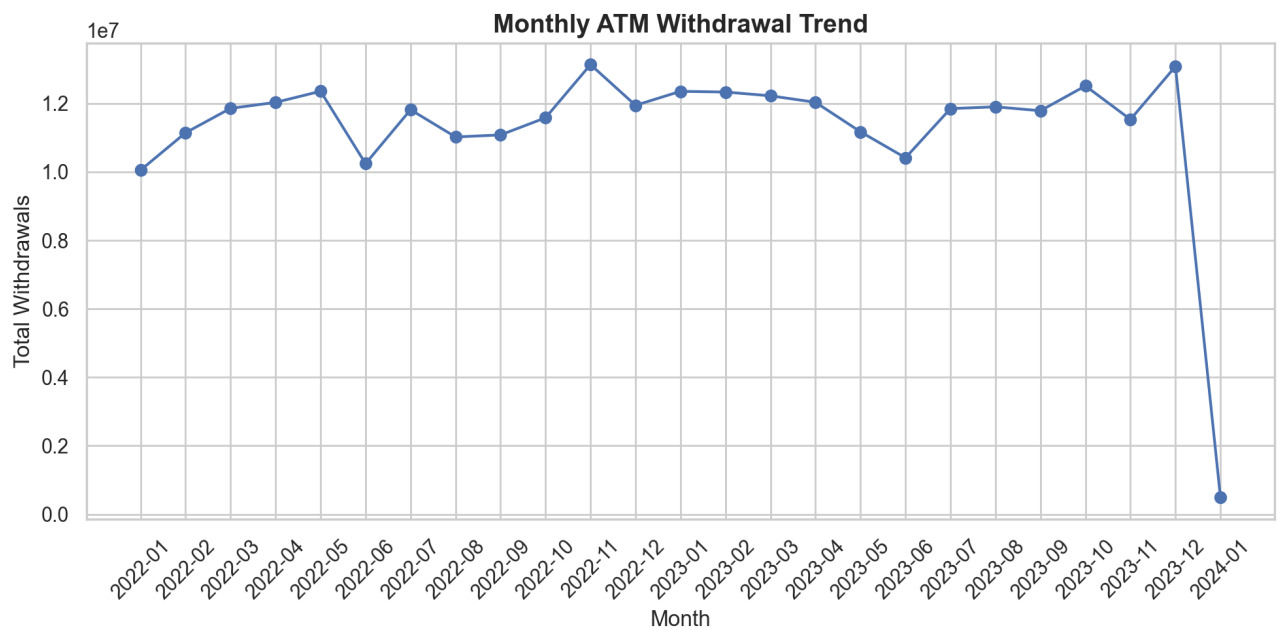
Attribute	Value
Records	5,658
Distinct ATMs	50
Date Range	2022-01-01 to 2024-01-01
Total Withdrawals	281,814,269
Total Deposits	57,310,326

2. Key Performance Indicators

Metric	Value
Total Records	5,658
Distinct ATMs	50
Total Withdrawals	281,814,269
Total Deposits	57,310,326
Average Withdrawal	49,808.11
Average Next-Day Demand	43,085.21
Average Utilization	52.05%
Withdrawal / Deposit Ratio	4.92
High-Risk Records	363

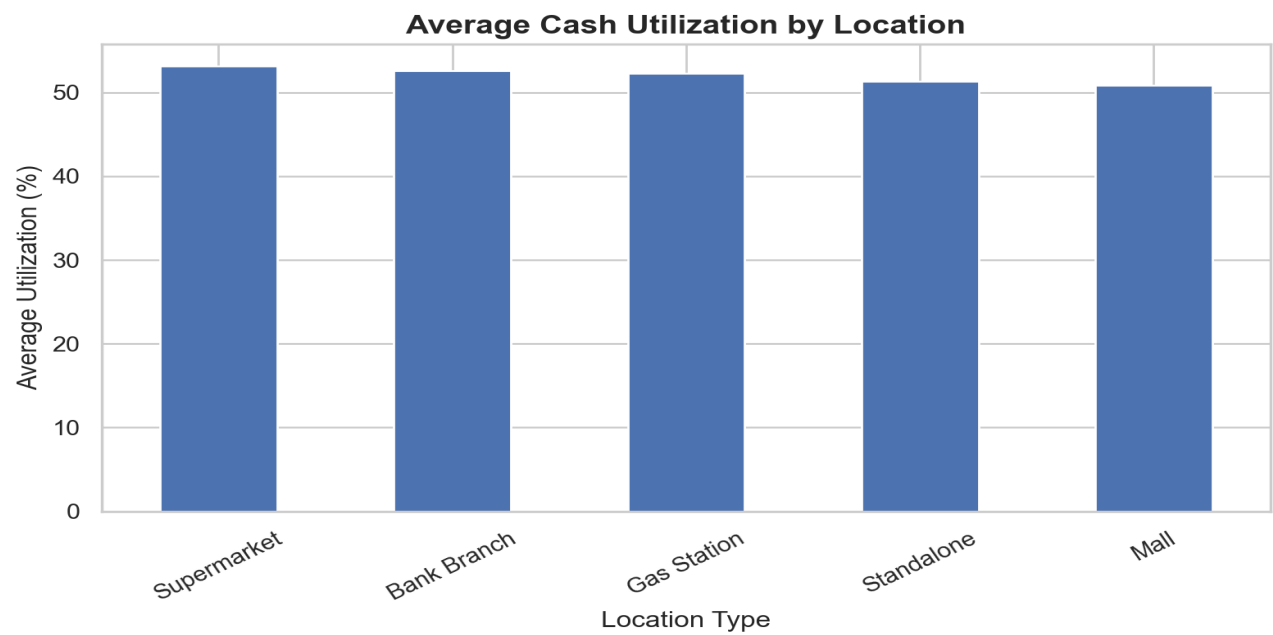
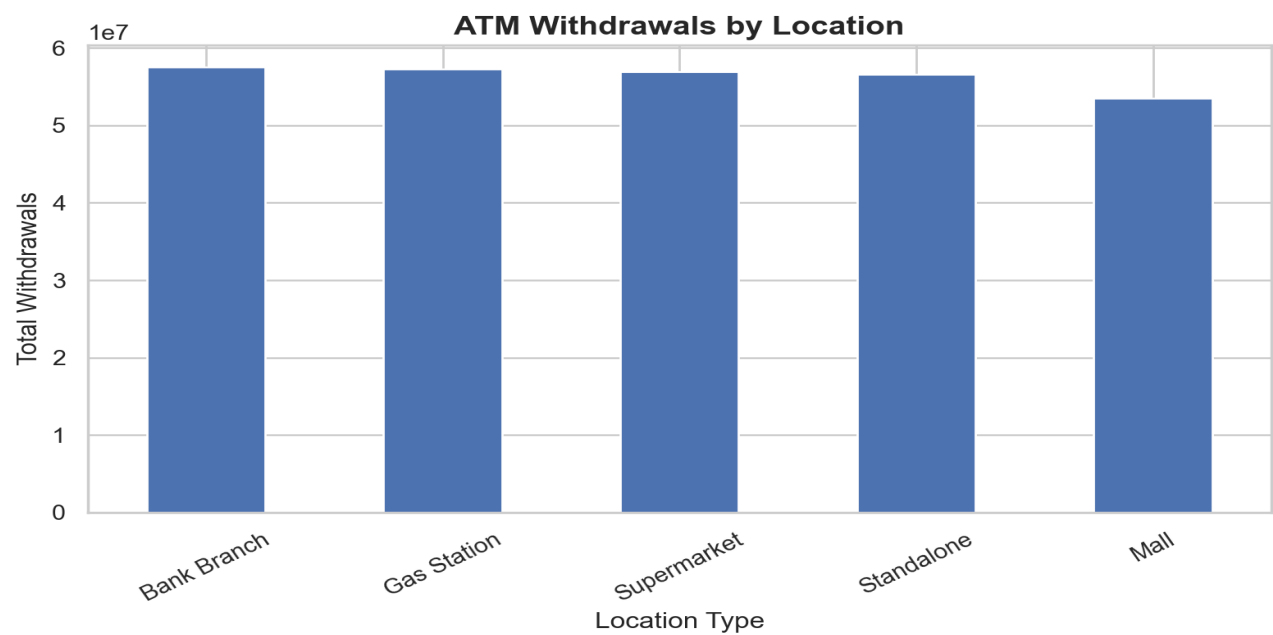
3. Monthly Withdrawal Trend

The monthly trend illustrates how aggregate ATM withdrawal activity changes over the analyzed period.



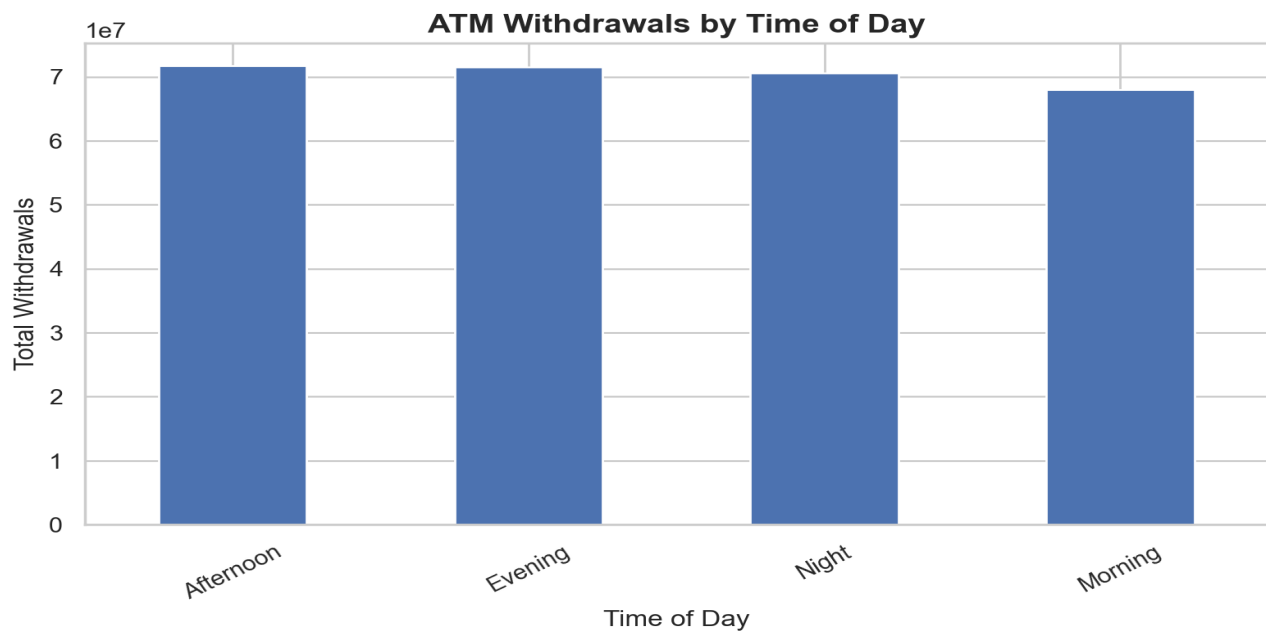
4. Location Analysis

Withdrawal activity is compared across the available ATM location categories to identify differences in transaction volume.



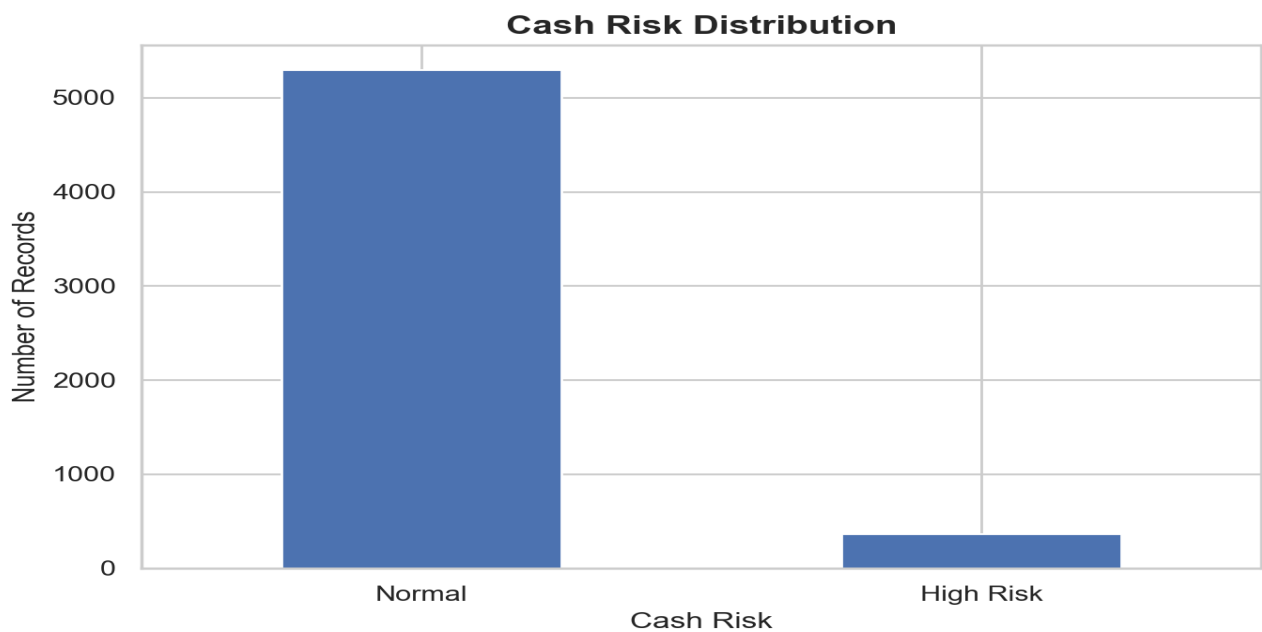
5. Time-Based Analysis

Time-of-day and day-of-week patterns provide an understanding of when ATM withdrawal activity is relatively higher or lower.



6. Cash Availability Risk Analysis

The Cash_Risk indicator is used to identify records where the combination of previous-day cash level and next-day cash demand indicates potential cash availability pressure.



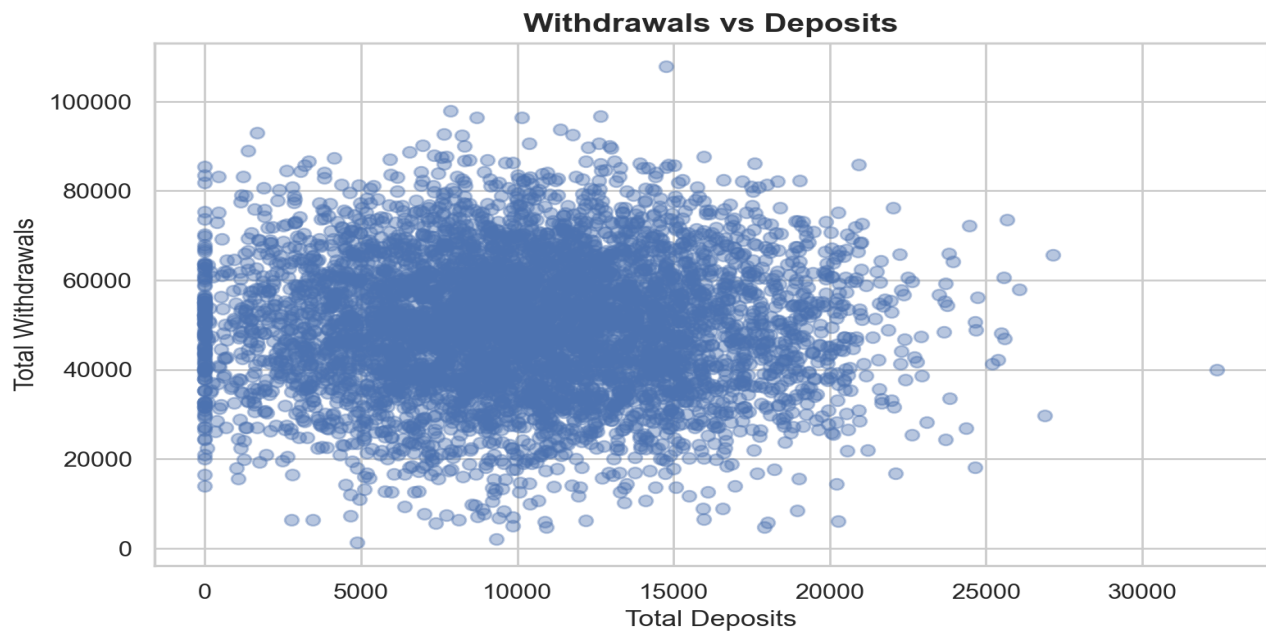
Important limitation: The dataset does not contain a direct ATM downtime or outage field. Therefore, Cash_Risk should be interpreted as a potential cash-availability risk indicator rather than confirmed ATM downtime.

7. Business Insights

1. The Bank Branch location category records the highest aggregate withdrawals with approximately 57,507,375 withdrawal transactions.
2. The highest average next-day cash demand is observed in the Bank Branch location category, at approximately 43,617.28.
3. The highest average cash utilization is observed in the Supermarket location category at 53.12%.
4. Afternoon records the highest aggregate withdrawal activity among the available time-of-day categories.
5. Thursday records the highest aggregate withdrawal activity among the available days of the week.
6. The dataset contains 363 records classified as High Risk, representing approximately 6.42% of all records.
7. Average cash utilization across the dataset is 52.05%.

8. Withdrawal and Deposit Relationship

The following visualization compares withdrawal and deposit activity across the dataset.



9. Conclusion

The analysis provides a structured view of ATM cash utilization, transaction behavior, cash demand and potential cash availability risk. The results can support data-driven monitoring of ATM activity and cash planning.

The final Power BI dashboard can be used as the interactive visualization layer for presenting these findings to business users.